# Ekaansh Sawaria

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## Summary

Backend and AI Engineer focused on **Agentic AI, RAG, LLM systems, and scalable backend infrastructure**, with experience in **multi-agent orchestration, deterministic decision systems, payment APIs, distributed systems, and system design**.

## Education

**Manipal University Jaipur** – B.Tech in Computer Science — CGPA: 8.51/10 *Aug 2023 – Present*

**Hiranandani Foundation School** – Class XII (CBSE) — 81% *2023*

**Hiranandani Foundation School** – Class X (CBSE) — 96% *2021*

## Experience

**Associate Software Engineering Intern** | *Accenture* *May 2026 – July 2026*

- Led and collaborated with a team of **3 interns** to architect an enterprise-scale **multi-agent RAG platform**, defining system architecture, retrieval workflows, and agent orchestration.
- Engineered a **Grok Build 0.1** agentic framework with multiple AI agents and **6 dynamically callable tools** for context-aware execution.
- Integrated **PageIndexer, BM25, dense retrieval, Cross-Encoder reranking, and QuadRank** into a hybrid search engine for high-precision enterprise product discovery.

## Projects

**RExplain – AI-Powered GitHub Repository Analysis Platform** *demo* — *github*

- Reduced repository analysis latency by **95%**, from minutes to **under 5 seconds**, using GitHub Tree API with selective retrieval across **8 critical file types**.
- Achieved **80% faster repeat queries** with a persistent **PostgreSQL (Neon)** embedding cache and SHA-based invalidation, eliminating re-indexing for **100% of unchanged repos**.
- Engineered an end-to-end **RAG pipeline** with SentenceTransformers, semantic retrieval, Groq LLM, FastAPI, React, PostgreSQL, and Docker.

**NEXORA – Agentic AI Commerce & Payment Infrastructure** *demo* — *github*

- Architected a **multi-agent commerce system** using LangGraph where AI buyer and merchant agents conduct multi-round negotiations across price, quantity, delivery, and warranty.
- Enforced financial boundaries through a **deterministic policy engine**, separating LLM reasoning from authorization and validating buyer constraints and merchant policy floors before payment execution.
- Secured the payment lifecycle with **Razorpay Orders API, Checkout, HMAC-SHA256 webhooks, PostgreSQL, and FastAPI**, alongside immutable agreements and cryptographically verifiable audit trails.

**LLM Quantization Sensitivity Analysis** *github*

- Evaluated **6+ transformer layer types** across DistilBERT and Phi-2 using **8-bit, NF4, and FP4** quantization to measure layer-wise sensitivity.
- Developed a mixed-precision framework achieving up to **40% memory reduction** with **<2% perplexity degradation**, benchmarking **3 quantization schemes**.

## Skills

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|---|---|
| **Languages** | Python, Java, C++, JavaScript, SQL |
| **Backend** | FastAPI, Flask, REST APIs, Microservices, SQLAlchemy, Payment APIs |
| **AI/ML** | Agentic AI, LLMs, RAG, LangGraph, Transformers, PyTorch, TensorFlow, Scikit-Learn, SentenceTransformers |
| **Databases** | PostgreSQL, MySQL, Vector Databases |
| **DevOps** | Docker, Git, GitHub, CI/CD |
| **Core CS** | DSA, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems, System Design |

## Achievements & Certifications

- **Hackathon Finalist** – HackerzStreet 2025; recognized at Sparkathon 2025.
- **Open Source & Coding:** 450+ GitHub contributions, 220+ LeetCode problems, ∼1500 LeetCode rating (Top 35%).
- **Certifications:** Neural Networks & Deep Learning, Hyperparameter Tuning, Convolutional Neural Networks (DeepLearning.AI, Coursera); Deep Learning Badge (Credly).